ANALYTICAL CUMULATIVE EXAM
APRIL 27, 1991

(30 points) 1. Consider the following data collected by replicate analysis: 15.32, 15.46, 15.26, 15.20 and 15.30.

a. What is the median and what is the mean of this data set?

b. What is the average deviation, what is the standard deviation and what is the variance?

c. Knowing that t = 4.60 for 4 degrees of freedom at the 99 percentile level, calculate the range of values which has a 99% probability of including the true value.

(10 points) 2. Potassium iodate is slowly introduced to a solution that is 0.040 M in Ag^+ and 0.050 M in Ba^{2+}. Which iodate will precipitate first and what will be the concentration of it remaining in solution when the second iodate begins to precipitate? (For $AgIO_3$ $K_{sp} = 3.00 \times 10^{-8}$ and for $Ba(IO_3)_2$ $K_{sp} = 1.57 \times 10^{-9}$.)

(15 points) 3. Knowing that the solubility product of silver phosphate is 1.30×10^{-20} and that of silver chloride is 1.7×10^{-10}, calculate the equilibrium constant for the reaction

$$Ag_3PO_4\ (s) + 3\ Cl^- \rightarrow 3\ AgCl\ (s) + PO_4^{3-}$$

(15 points) 4. Calculate the pH of a solution prepared by mixing together

100 mL of 0.05 M HNO_3
50 mL of 0.03 M H_2SO_4
75 mL of 0.20 M NaOH
40 mL of 0.06 M $Ca(OH)_2$
100 mL of water

(15 points) 5. Calculate the pH of a solution prepared by mixing 100 mL of a 0.08 M solution of a weak acid ($pK_a = 4.5$) and 25 mL of a 0.10 M solution of NaOH.

(15 points) 6. Phosphoric acid has $K_{a1} = 7.11 \times 10^{-3}$, $K_{a2} = 6.34 \times 10^{-8}$, and $K_{a3} = 4.2 \times 10^{-13}$. Bromcresol green changes color in the pH range of 3.8 to 5.4 and thymolphthalein changes in the range 9.3 to 10.5. A sample which may contain HCl, H_3PO_4, and NaH_2PO_4 or any mixture of these three is titrated with a 0.117 N solution of NaOH. It requires 18.16 mL to reach the first color change and an additional 47.32 to reach the second color change. Which of the three components were present in the mixture and how many millimoles of each were there?